Targeting Sprint 2 - Counter-Air Evaluation - Categories 1, 2, 3 & 4 (Integrated MOSA C2 / Project AVO Interlock & 10 U.S.C. 4023 Payload)

Submitting Entity: Humphrey Virtual Farms LLC (701 E Drew, Madill, OK 73446-2620 | 100% U.S. Sovereign Small Business, 0% FOCI)

Statutory Identifiers: CAGE Code: 1AHA8 | SAM.gov UEI: S1M4ENLHTDH5 | Non-Traditional Defense Contractor (10 U.S.C. 4025 / 4022 / 4023)

Principal Investigator & CEO: Jeffery Humphrey | Phone: (572) 358-9978 | Email: humphreyvirtualfarm@gmail.com

Public Verification Repository: https://github.com/mrshumphrey3251-ai/hvf-media-matrix-public

Shared Core C2 Cryptographic Merkle Root (640 Compiled Python C2 Modules): d65049efcf06298f84d41436131794fc3cc5607873a2a596b5ad355f6ee67ccb

[+] Executive BLUF: Project Ebony v8.0.0 delivers a production-ready, software-defined Three-Brain MOSA Command-and-Control (C2), Project AVO Interoperability Bridge, and Fail-Safe Over-the-Horizon Human-in-the-Loop Weapon Safety Interlock Suite engineered for CAT 1 (AGM-114L/AGM-179 GFI TDP), CAT 2 (APKWS), CAT 3 (Stabilized RWS/Stinger/HPL), and CAT 4 Counter-Air USVs.

[+] Dual Contracting & Fielding Readiness: Available immediately (1) as an integrated C2/Safety Interlock subsystem teamed with CAT 1-4 USV hull and effector primes for 120-day forward operational fielding under 10 U.S.C. 4025 / 4022, and (2) for direct Government experimental procurement of C2/safety-interlock payloads and SIL/HIL test articles under 10 U.S.C. 4023 ($1,485,000.00 / 12-Month Fixed-Price Schedule; 100% Section 889 Compliant).

Verified 640-Module C2 Baseline, Active Software/Payload Production Line & Cross-Agency Federal Validation

[+] Demonstrated System Maturity: Project Ebony v8.0.0 is a fully compiled, deterministic 640-module Python C2 and SCADA/actuator interlock suite verified across 85/85 automated unit test suites, 17/17 blocked Red-Team cyber-physical penetration vectors (with automatic single-letter tamper zeroization), and 52 bare-metal SQLite WORM/vault tables on local loopback (127.0.0.1) with 0% commercial cloud dependency.

[+] Cross-Referenced DoD & Federal Validation Pedigree: Active DoD CDAO Tradewinds Solutions Marketplace Docket 10-26-4046 (Status: Compliant/Queued for Assessment -- Oct 29, 2026), DIU Portfolio Network Intake (Status: Received -- Oct 31, 2026), U.S. Army DEVCOM ARL BAA W911NF-23-S-0001 ($750K White Paper Transmitted), USACE ERDC BAA ($1.2M), and DHS S&T LRBAA ($1.0M OIP Registered).

[+] Active Production Line & Surge Capacity: Zero custom fab lead-time bottlenecks. Project Ebony v8.0.0 deploys as a containerized, cryptographically sealed bare-metal image onto commercial off-the-shelf (COTS) ruggedized marine compute nodes and standard RS-485/CAN-bus/SiC MOSFET solid-state interlocks, enabling 100+ unit/month software/firmware replication and immediate 30-day Test Event 1 deployment.

[+] Baseline Hull & Effector Integration Agnosticism: Interfaces natively with any >=200 NM range, >=10 knot USV hull and CAT 1-4 launcher via standardized RS-485 Modbus RTU, CAN-bus (J1939/NMEA 2000), and MIL-STD-1760 / GFI TDP electrical fire-control relays.

Hardware-Enforced Separation of Weapon Custody (Brain 1), Clutter-Resilient PID (Brain 2) & WORM Audit (Brain 3)

[+] Brain One (Deterministic Effector Safety Interlock & Human-in-the-Loop Custody Core): Executes a <50 microsecond bare-metal safety loop governing over-the-horizon remote arming, firing, safe/de-arming, and mission abort across CAT 1-4 launchers. Enforces a 2.04 microsecond RS-485 Modbus RTU (FC05) emergency actuator cutoff and a 36.75 nanosecond Silicon Carbide (SiC) MOSFET gate isolation target (1,360,544x faster than a 50ms marine PLC).

[+] Fail-Safe Stabilization & Launch Envelope Enforcement: Brain One continuously monitors vessel pitch, roll, yaw rate, watchdog heartbeat, and authenticated operator custody tokens; if wave motion exceeds safe launcher gimbal limits or operator abort is commanded, Brain One physically severs firing bus power in <50 microseconds.

[+] Brain Two (360-Degree Clutter-Resilient Sensor Fusion & DDIL Analyst): Executes zero-cloud edge neural inference for 360-degree EO/IR, radar, and acoustic detection, tracking, and positive identification (PID) of Group 1-5 UAS, rotary-wing aircraft, and surface threats at >3 NM in high sea clutter. Communicates with Brain One strictly across a unidirectional read-only memory diode with zero physical wire to launcher firing circuits.

[+] Brain Three (Sub-Millisecond WORM Merkle DAG Provenance Arbiter): Seals every tactical cue, target track, human-in-the-loop arm/fire/abort state, and kinematic telemetry frame into a 52-table local SQLite WORM Merkle DAG in <0.2 milliseconds per event.

Turnkey Project AVO SDK/API Interoperability & 3DOF/6DOF Software/Hardware-in-the-Loop Simulation Suite

[+] Seamless Project AVO (Autonomous Vehicle Orchestrator) Integration: Built on Modular Open Systems Approach (MOSA) principles with a documented, schema-validated JSON/REST, gRPC, and RS-485/CAN-bus C2 Interface Technical Data Package (SDK/API) ready prior to Test Event 1 and Test Event 2 (Little Creek, VA).

[+] Offboard Tactical Cueing & Supervised Remote Engagement: Securely ingests offboard target coordinates from Government-directed air/surface sensors via Project AVO, computes intercept solutions, slews CAT 1-4 launchers onto target bearing, and presents cryptographic human-in-the-loop arm/fire/abort controls with minimal operator workload.

[+] Built-In SIL/HIL Kinematic & Error-Condition Verification: Includes a deterministic Software-in-the-Loop (SIL) and Hardware-in-the-Loop (HIL) simulation harness modeling 3DOF/6DOF target and effector envelopes (>3 NM range, Group 1-5 UAS & rotary-wing profiles, vessel sea-state pitch/roll dynamics, and launcher fault/abort injection) for immediate Test Event 1 engineering deposition.

[+] Cryptographic Anti-Tamper & Zero-Trust C2 Security: All C2 links and onboard modules are protected by SHA-256 Merkle DAG verification; any unauthorized byte modification or replay injection triggers instant interlock safe/de-arming and enclave zeroization (17/17 Red-Team vectors blocked).

Modular Fire-Control & Safety Interlock Compatibility Across All Four SWAP-USV Sprint 2 Capability Tracks

[+] Category 1 (Heavy Kinetic -- AGM-114L Longbow Hellfire / AGM-179 JAGM): Ready to ingest the Government Furnished Information (GFI) Technical Data Package (TDP) under CRADA (Active CAGE: 1AHA8) to bind Brain One’s <50us human-in-the-loop safety interlock and Project AVO telemetry bridge directly to the GFI heavy kinetic launcher interface.

[+] Category 2 (Precision Rocketry -- APKWS): Provides rapid multi-target re-engagement sequencing, laser-designation safety interlock gating, and two-person pier-side/austere reload safe/de-arming lockout verification for 70mm APKWS launchers against Group 1-3 UAS swarms.

[+] Category 3 (Remote Weapon Stations -- Stinger / 7.62mm+ RWS / High-Power Laser): Provides high-speed horizon-stabilization cueing handoff, firing-cutout zone enforcement (protecting own-ship superstructure and friendly vessels), and 24V DC / SiC solid-state power management for kinetic RWS and directed-energy HPL effectors.

[+] Category 4 (Alternate Mature Kinetic Launchers & Offensive Surface Mode): Supports modular tube/box launcher rearming interlocks and dual-mode Counter-Air / Offensive Surface engagement profiles against explosive USVs, go-fast attack craft, and littoral UAS launch sites.

28.5W / 24V DC Auxiliary Footprint, GNSS-Denied Resilience & Turnkey 20/40ft ISO Container Transportability

[+] DDIL & GNSS-Denied Autonomous Localization: Operates continuously over full 200+ NM loiter missions with 0% commercial cloud or external internet dependency. When GNSS is jammed, degraded, or spoofed, Brain Two fuses optical horizon/landmark bearings, inertial dead-reckoning, and RF beacon angle-of-arrival to maintain waypoint navigation, station-keeping loiter, and collision avoidance.

[+] Ultra-Low SWaP-C Shipboard Footprint: Draws only 28.5W continuous auxiliary power on native 24V DC marine electrical buses (eliminating high-kilowatt server racks and liquid cooling), preserving USV fuel endurance and payload weight capacity for maximum counter-air magazine depth.

[+] Zero-MHE Turnkey Logistics (C-130 / C-17 / 20ft & 40ft ISO Container / Well-Deck Ready): Compact, sealed solid-state enclosure mounts directly inside standard USV electronics bays or 20/40ft ISO containerized launchers, requiring zero specialized Material Handling Equipment (MHE) or pre-mission calibration.

[+] Two-Person Pier-Side & Shipboard Rearming Safety Lockout: Physical + cryptographic deck-maintenance lockout key ensures 100% electrically dead firing buses (<36.75ns SiC MOSFET isolation) during two-person manual reload operations in rough littoral states.

Turnkey 30-Day Demonstration Readiness & 120-Day Forward Operational Fielding Schedule

[+] Day 1 - 15 (CRADA, Category 1 GFI TDP & USV Hull/Launcher Binding): Execute Category 1 GFI TDP CRADA (CAGE: 1AHA8), bind Brain One RS-485/CAN-bus/MIL-STD fire-control interlocks to partner USV & CAT 1-4 launcher hardware, and validate 3DOF/6DOF SIL/HIL kinematic models.

[+] Day 16 - 30 (Test Event 1 -- Onsite Baseline Integration, Safety & Dry-Fire Deposition): Execute 1-2 day live demonstration of LOS/BLOS autonomous maneuvering, 360-degree clutter-resilient target tracking, human-in-the-loop remote arming/firing/de-arming/abort interlocks, dry-fire launcher cutoff (<50us), and full SIL/HIL engineering deposition.

[+] Day 31 - 75 (Test Event 2 -- Little Creek, VA Operational Challenge): Deliver full Project AVO C2 integration TDP (SDK/API & emulator), execute DoW-operated unified C2 control, DDIL/GNSS-denied transit under active jamming, offboard tactical cueing, and live/simulated counter-air engagement against Group 1-5 UAS and rotary-wing targets.

[+] Day 76 - 120 (Forward Operational Fielding & Scaled Production): Deliver production-sealed Three-Brain C2 & Safety Interlock kits across fleet USVs under 10 U.S.C. 4025 / 4022 / 4023.

Favorable Cost-Exchange Ratio, 100% Background IP Retention & Immediate 10 U.S.C. 4025 / 4022 / 4023 Awardability

[+] Shifting the Maritime Cost-Exchange Ratio: Replaces brittle, multi-million-dollar proprietary naval combat management suites with a $1,485,000.00 Fixed-Price Phase-1 Prototype OTA / 10 U.S.C. 4023 Experimental Subsystem package ($297,000.00 Day 15-30 Mobilization | 0% Cost Share) and <$15,000 per-vessel recurring COTS hardware/interlock kit cost at scale.

[+] Intellectual Property & Government Purpose Rights: Humphrey Virtual Farms LLC retains 100% of existing background IP in Project Ebony v8.0.0 (DFARS 252.227-7018) while granting DIU and DoW the Challenge evaluation license and open MOSA C2 / Project AVO interface rights.

[+] Immediate Teaming & Standalone 10 U.S.C. 4023 Procurement: Ready for immediate pairing by DIU/DRPM UxS with any down-selected CAT 1-4 USV hull/effector prime OR direct non-competitive experimental procurement under 10 U.S.C. 4023 as DoW’s standardized USV Counter-Air Safety Interlock, WORM Flight-Data Recorder, and Project AVO SIL/HIL Test Article.

[+] Cryptographic Attestation: Verified across 640 compiled Python C2 modules under Shared Core C2 Merkle Root d65049efcf06298f84d41436131794fc3cc5607873a2a596b5ad355f6ee67ccb (CAGE: 1AHA8 | SAM.gov UEI: S1M4ENLHTDH5 | 100% Section 889 NDAA Compliant).